

Cybersecurity framework for connected and automated vehicles: A modelling perspective

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ABSTRACT

Connected and Automated Vehicles (CAVs) cybersecurity is an inherently complex, multi-dimensional issue that goes beyond isolated hardware or software vulnerabilities, extending to human threats, network vulnerabilities, and broader system-level risks. Currently, no formal, comprehensive tool exists that integrates these diverse dimensions into a unified framework for CAV cybersecurity assessment. This study addresses this challenge by developing a System Dynamics (SD) model for strategic cybersecurity assessment that considers technological challenges, human threats, and public cybersecurity awareness during the CAV rollout. Specifically, the model incorporates a novel SD-based Stock-and-Flow Model (SFM) that maps six key parameters influencing cyberattacks at the system level. These parameters include CAV communication safety, user adoption rates, log file management, hacker capabilities, understanding of hacker motivations (criminology theory maturity), and public awareness of CAV cybersecurity.

The SFM's structure and behaviour were rigorously tested and then used to analyse five plausible scenarios: i) Baseline (Technological Focus Only), ii) Understanding Hacker Motivations, iii) CAV User and OEM Education, iv) CAV Penetration Rate Increase, and v) CAV Penetration Rate Increase with Human behaviour Analysis. Four metrics are used to benchmark CAV cybersecurity: communication safety, probability of hacking attempts, probability of successful defence, and number of CAV adopters. The results indicate that while baseline technological advancements strengthen communication framework robustness, they may also create new vulnerabilities that hackers could exploit. Conversely, a deeper understanding of hacker motivations (Criminology Theory Maturity) effectively reduces hacking attempts. It fosters a more secure environment for early CAV adopters. Additionally, educating CAV users and OEM increases the probability of defending against cyberattacks. While CAV penetration increases the likelihood of hack defence due to a corresponding rise in attempts, there is a noticeable decrease in hacking attempts with CAV penetration when analysing human behaviour. These findings, when translated into policy instruments, can pave the way for a more optimised and resilient cyber-safe ITS.

1. Introduction

Deploying Connected and Automated Vehicles (CAVs) is a unique opportunity for the transport industry's vendors, regulators, and customers to reassess the understanding, interpretation, and architecture of Intelligent Transportation Systems (ITS). Indeed, CAVs would revolutionise many of the inherent dynamics governing transport systems by streamlining and optimising their deployment. The key driver behind these innovations is pervasive external and internal connectivity (data, controls, and commands). This digital transformation has resulted in new partnerships between technology companies and traditional automakers, extending illicit activities in both the cyber and physical spheres. The majority of car network protocols were developed prior to automobiles being connected and therefore lack cybersecurity

measures.

There is no general characterisation of a cybersecurity breach nor a definitive list of threats. Sophisticated cybersecurity attacks often do not have an end. The challenge is to safeguard CAVs against all types of cyber threats. Any single incident or sequence of events that imperil the availability, reliability, confidentiality, authentication, robustness, trustworthiness, and integrity of CAV's digital information could lead to unexpected attack scenarios, posing severe risks to infrastructure and CAV users (Khan et al., 2020).

A successful cyber-attack on CAV has two major repercussions: an invasion of privacy and compromised safety. Safety hazards include system malfunctions, incorrect speed and location values, failure in object detection, traffic congestion, deceptive toxic signs, and compromised longitudinal safety and vehicle control (Yan et al., 2016). Privacy

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concerns arise on three fronts. First, at the consumer level: autonomy and surveillance of CAV users. Secondly, proximity espionage occurs through monitoring individuals in the vicinity by CAV sensors (radar, lidar, laser, camera), resulting in a significant invasion of privacy. The monitoring encompasses a comprehensive 360-degree observation of body movement and bio-signalling data, enabling the identification of individuals (Miller et al., 2020). For example, citing concerns about sensor spying, the Chinese military banned Tesla cars (Aljazeera, 2021). Thirdly, the vendor's intellectual property is at risk, as seen in cases like the guilty plea of an individual involved in a Tesla ransomware plot in the United States (BBC, 2021).

Security researchers have demonstrated that the Nissan Leaf is vulnerable to remote hacking due to weaknesses in the vehicle's Application Programming Interfaces (APIs) provided by the manufacturer. By exploiting these APIs alongside the easily accessible Vehicle Identification Number (VIN), attackers could potentially gain control over features such as climate control and heated seating (Coynne, 2016). Similarly, Fiat Chrysler issued a safety recall affecting 1.4 million vehicles in the US after security researchers revealed that one of its cars could be hacked (BBC, 2015).

Software vulnerability in the Tesla AV precipitated leakage of sensitive data (Prevost and Kettani, 2019). Vulnerabilities within the routing table and non-encrypted communications resulted in erroneous GPS positioning in the Google vehicle (Hirz and Walzel, 2018). The Mitsubishi Outlander experienced a breach through an insecure messaging protocol, leading to arbitrary light activation, deactivation, and disabling of anti-theft alarms (Jagielski et al., 2018). In a similar incident, hackers integrated spyware into the on-board diagnostic parts, disabling the BMW vehicle's diagnostic system (Tan et al., 2017). Therefore, cybersecurity for CAVs is essential for mitigating cyberattacks and deterring safety and privacy implications.

1.1. The contribution and scope of the study

The cybersecurity of CAVs poses a complex, multifaceted challenge that extends beyond hardware or software integration. It encompasses a range of interconnected factors across all CAV-based Intelligent Transportation Systems (ITS) elements. A comprehensive tool for system-level cybersecurity risk assessment remains absent. Existing models and mitigation methods typically focus on isolated aspects or specific scenarios, failing to capture the broader, interconnected nature of CAVs within ITS. This study addresses this challenge by developing an SD model for strategic cybersecurity assessment that considers technological challenges, human threats, and public cybersecurity awareness during the CAV rollout. Specifically, the model incorporates a novel SD-based Stock-and-Flow Model (SFM) that maps six key parameters influencing cyberattacks at the system level. These parameters include CAV communication safety, user adoption rates, log file management, hacker capabilities, understanding of hacker motivations (criminology theory maturity), and public awareness of CAV cybersecurity.

While SD is an established modelling approach, its application to this strategic cybersecurity challenge within the context of CAVs is novel. By examining these interactions, our model offers valuable insights into the emergent nature of cybersecurity risks, the long-term indirect effects of cyber threats, and the role of policy resistance. Furthermore, it provides actionable recommendations for vendors and regulators, including the need for enhanced automated protections, robust public awareness programs, and systematic policies for managing data logs—areas that have been underexplored in existing models.

This study highlights the critical role of proactive, system-wide risk management strategies as CAV deployment accelerates, increasing exposure to potential cyberattacks. Our model identifies technological and human vulnerabilities and emphasises the importance of mitigating these risks through an integrated, dynamic approach that can evolve alongside CAV adoption.

1.2. Structure and guide to the acronym

The following section discusses approaches to assessing the cybersecurity of CAVs. Section 3 outlines the methodology adopted, elaborating on the dynamic aspects of CAV's cybersecurity and introducing SD modelling and its strengths for evaluating CAVs' cybersecurity. Then, in Section 4, we described the structure of SFM, i.e., the scope, significance, relevance, and impact of each factor in the cybersecurity paradigm of CAVs. Model testing is the focus of Section 5. Furthermore, Section 6 discusses the findings of the simulations, followed by a discussion and policy recommendations. Finally, in Section 8, the limitations and future extensions are described.

Table 1 lists the acronyms used in this study.

2. Overview of CAVs cybersecurity assessment approaches

Several academic and industry experts have worked to synthesise the cybersecurity of automobiles. Typically, two methods may be employed: simulating cyberattacks or cyber threats on an operational testbed or on a live system (Van et al., 2010; Veksler et al., 2018). While CAV testing on public roads is still in its developmental stages globally, companies such as Waymo and Cruise have launched self-driving taxis for public use in select locations in the USA, including Arizona and San Francisco (Chari, 2024). However, large-scale deployment is still to be done. Therefore, the latter approach is challenging to deploy and unworkable. Similarly, the former required enhanced data access and collection, robust behavioural simulation models (Kavak et al., 2021), and security specialists to create well-designed cyber-attack scenarios (Yoo et al., 2020).

Schmidt et al. (Schmidt et al., 2014) proposed a security analysis technique for identifying and prioritising security vulnerabilities. However, the study takes an analytical approach to networked connections. It lacks an inter-avenue assessment, such as a concurrent assessment of the technological impact on CAVs' cybersecurity and hackers' capability. Similarly, Ward et al. (Ward et al., 2013) described an approach for assessing security risks in the automotive sector: threat analysis and risk assessment based on the HARA. Likewise, the authors (Macher et al., 2015) proposed a hybrid strategy, SAHARA, combining the automotive HARA and security domain STRIDE approaches, which elaborates on the influence of security concerns on system-level safety ideas. However, the study lacks a simultaneous assessment of various parameters from different avenues, such as the impact of CAV data files

Table 1
Abbreviation list.

Abbreviation	Explanation
ABM	Agent-Based Modelling
APIs	Application Programming Interfaces
BDM	Bass Diffusion Mode
CAVs	Connected and Automated Vehicles
CAVs-CS	CAVs Communication Safety
CMT	Criminology Theory Maturity
CV	Conventional Vehicle
DES	Discrete Event Simulation
eSTU	eSafety Traffic Unit
HARA	Hazard Analysis and Risk Assessment
ITS	Intelligent Transportation Systems
MCS	Monte Carlo Simulation
NM	Network Modelling
PHA	Probability of Hacking Attempts
PHD	Probability of Hacks Defended
SD	System Dynamics
SFM	Stock-and-Flow Model
STRIDE	Spoofing, Tampering, Repudiation, Information disclosure, Denial of service and Elevation of privilege
TM	Technological Maturation
TRM	Technology Readiness and Maturity
V2X	Vehicle-to-Everything

on hackers' capabilities as well as an input to CAV technological advancement.

Burzio et al. (Burzio et al., 2018) presented a ranking-based high-level description of CAVs cybersecurity certification by two approaches: i) vehicle safety, the Euro New Car Assessment Programme assessment, and ii) a proposed cybersecurity ranking approach: assessment of the components, subsystems, and topology of the connection will result in a ranking of the overall system. However, the paper only covers a high-level hierarchy from a cyber security perspective in the automotive industry and lacks focused methods for covering this issue in-depth. The authors (Raiyn, 2018) suggested a cyber-attack defence paradigm based on iris recognition that utilises biometric data for message authentication and safe transmission. Nevertheless, the study is focused on flow-charts and is based on a single avenue of analysis—CAVs communication.

Sheehan et al. (Sheehan et al., 2019) proposed a CAV cyber-risk classification model by encompassing software vulnerabilities in the United States Vulnerability Database. However, this is singleOnarorwd focus data base and the model's validity is still to be tested for ranking and classifying cyberattacks in the ITS. Khan et al. (Khan et al., 2023a) have recently conducted a dynamic assessment of regulation and CAVs adoption; however, the study primarily focuses on cyber regulation. Furthermore, there is a paucity of concurrent research into the interconnected characteristics of ITS components. Understanding this is crucial as CAVs will navigate dynamic environments, presenting substantial technical hurdles and escalating system complexity.

3. Methodology

The most difficult challenge is developing and implementing a robust cybersecurity system for CAVs. Addressing this entails numerous inter-linked factors of diverse nature, absent of empirical data. Developing future insights amid such uncertainty is an alternate strategy. We have chosen SFM: an approach used to analyses complex, interrelated, and uncertain CAV technology and to explore the cybersecurity implications of CAVs on system-level (Stermann, 2000; Stasinopoulos et al., 2020; Khan et al., 2024a, 2024b). SD is employed to evaluate the cybersecurity of Level 4 (and above) CAVs adhering to a functional pathway. The variables and their correlation in the model are based on established innovation theory, a meta-investigative quantitative review of the literature (post-2010), sourced from a various sources, including books, academic databases, industry reports and PhD theses, supported by forward and backward snowballing (Hidalgo and Albors, 2008). This culminated in recognising key domains at the heart of CAVs cyber-security study: CAVs *Communication Safety* (CAVs-CS), CAV *Adopters*, *Hacker's Capability*, *Log files*, *Criminology Theory Maturity* (CMT), and *Public CAVs Cyber Safety Awareness*.

To benchmark and evaluate cybersecurity in CAVs across different scenarios, four measures are used: i) CAV *adopters*, ii) PHA (*Probability of Hacking Attempts*), CAVs-CS, and iv) PHD (*Probability of Hacks Defended*). The CAVs-CS serves as the primary defense against cyber threats and demonstrates the increased resilience of technological advancement. The PHD is important to investigate because a significant portion of defended hacks reduces the prevalence of successful cyberattacks. In addition, the PHA is crucial to synthesise because an increase in hacking attempts indicates an increase in successful breaches of CAV connectivity, which hinders the adoption of CAVs.

The work on cyber breaches against CAVs (Khan et al., 2020), cybersecurity evaluation frameworks for CAVs (Khan et al., 2021a, 2023a), and empirical examination of perceived cyber risks to CAV rollout (Khan et al., 2023b, 2024c) enabled to reconstruct the interrelations of various cybersecurity factors and served as the foundation for the present study. Other aspects, such as regulatory laws or trust, are beyond the scope of this study. The selection is primarily motivated by the need for a restricted boundary for assessing a small number of variables and the non-geographic nature of the SD model; however, these aspects warrant further study.

3.1. The rationale for using SD-based SFM

CAV technology is evolving, enabling dynamic complexities. It has the characteristics of paradoxical behaviour: the various components that facilitate CAV's cyber protection are interlinked, feedback-driven, and non-linear; a change in a single parameter triggers further behaviour, leading to a new scenario that affects subsequent decisions (Khan et al., 2021a). Secondly, the cybersecurity of CAVs can be linked to a chain, wherein the efficacy of the entire link is contingent upon the robustness of its accompanying links. The robustness of technological components, human threats, and consumer cyber-behaviour are just a few of the links in the chain that make up CAV's cybersecurity (Sarriegi et al., 2006). "Nested complexity" exists when a well-organised governing system (, i.e., ITS) and regulatory framework (regulators/policy makers) regulate a physical system (CAV) that is confronted with contemporary challenges. An unconditional, technology-driven deployment of CAVs might conflict with social and environmental sustainability goals. Moreover, uncertainty and technical concerns are incorporated into CAV technology (Tan and Taeihagh, 2021).

Dynamic facets of CAVs cybersecurity could be investigated by Agent-Based Modelling, simulation-based models (Nazareth and Choi, 2015), queueing theory, risk assessment and threat analysis (Ward et al., 2013), or security domain STRIDE approaches (Macher et al., 2015). However, these unidirectional models are incapable of explaining paradoxical behaviour, are difficult to apply to large and complex systems, lack inter-avenue feedback, and are inadequate in design situations where data is scarce. Therefore, to model the history-dependent, adaptive, counter-intuitive, and policy-resistant characteristics of CAVs cybersecurity, SFM provides a valuable framework for comprehending and understanding challenges. Indeed, SFM facilitates the exploration of potential scenarios, fostering the development of a unified array of potent cybersecurity strategies aimed at addressing uncertain CAV technological challenges.

SD approaches, developed in the early 1960s, provide a solid theoretical foundation for understanding the operation of complex systems (Stermann, 2000). Notably, SD approaches have been employed to scrutinise and evaluate a broad range of complex systems, encompassing information security (Nazareth and Choi, 2015), cybersecurity (Sarriegi et al., 2006), and highway congestion (Goodman, 1974). Within the CAV literature, the application of the SD approach predominantly centres on the integration of driverless cars into ITS (Stanford, 2015). By employing the SD approach, Gruel and Stanford (2016) assessed the long-term consequences of automated vehicles, while Nieuwenhuijsen et al. (Nieuwenhuijsen et al., 2018) scrutinised the long-term diffusion of automated vehicle technology. In a similar vein, Puylaert et al. (Puylaert et al., 2018) examined preliminary forms of automated driving using SD methodology. Furthermore, Stasinopoulos et al. (Stasinopoulos et al., 2020) explored the repercussions of CAV adoption on greenhouse gas emissions.

Table 2 compares SD and other potential modelling approaches, such as Agent-Based Modelling, Discrete Event Simulation, Network Modelling, Monte Carlo Simulation, Hazard Analysis and Risk Assessment, and STRIDE. This comparison illustrates SD is particularly suited for system-level cybersecurity assessments in CAVs over these other approaches.

4. The development of the SFM

In the realm of system dynamics, two pivotal concepts are the "system" itself and the notion of "system change" (Stermann, 2000). A system is a cohesive whole consisting of various constituent elements that operate in conjunction to fulfil functions that may not be readily discernible from the functioning of individual components (Hirsch et al., 2007). The CAV-based ITS system requires the collaboration of multiple stakeholders (automakers, communication service providers, and consumers). Notably, the seamless operation of CAVs heavily relies on interactive coordination and collaboration with ITS communication

Table 2
SD vs Different Model comparison for CAVs cybersecurity assessment.

Aspect	System Dynamics (SD) (Sterman, 2000 ; Khan et al., 2022)	Agent-Based Modelling (ABM) (Macal and North, 2005)	Discrete Event Simulation (DES) (Günel and Pidd, 2010)	Network Modelling (NM) (Smith, 2013)	Monte Carlo Simulation (MCS) (Fagade et al., 2017)	Hazard Analysis and Risk Assessment (HARA) (Macher et al., 2015)	Spoofing, Tampering, Repudiation, Information disclosure, Denial of service and Elevation of privilege (STRIDE) (Abuabed et al., 2023)
Approach	Continuous	Discontinuous	Discrete	Discrete or Continuous	Stochastic, Random Sampling	Risk-Based, Structured	Threat-Based, Structured
Level Perspective	Macroscopic Aggregated (e.g., overall cybersecurity trends)	Microscopic Disaggregated (e.g., individual vehicle behaviours)	Both (Micro/Macro) Process-Oriented (e.g., event sequences in attacks)	Microscopic or Mesoscopic Structural Relationships (e.g., network topologies)	Macroscopic Probabilistic Outcomes (e.g., likelihood of breaches)	System-Level Safety and Risk Evaluation	Component/Threat-Level Identifying Threats across System Components
Central Concept	Feedback loops (e.g., cybersecurity policies, threat escalation)	Objectives, rules (e.g., attacker and defender strategies)	Events and Resources (e.g., intrusion detection, response times)	Nodes, Edges (e.g., V2X communication paths, attack vectors)	Probability Distributions (e.g., attack success rates)	Potential Hazards (e.g., V2X security, vehicle control systems)	Attack Vectors (e.g., communication protocols, system interfaces)
System Components	Stocks and flows (e.g., attack surfaces, threat levels)	Agents and relations (e.g., vehicles, attackers, defence systems)	Entities, Events (e.g., cybersecurity incidents, response protocols)	Nodes, Links (e.g., CAVs, communication channels)	Variables, Probabilistic Models (e.g., system vulnerabilities)	Hazard Identification and Risk Control	Attack Tree Models, Threat Matrices
Simulation Engine	Integration of time steps (e.g., gradual evolution of threats)	Event-based scheduling (e.g., simulation of attack/defence scenarios)	Event-based scheduling (e.g., time-triggered cyber events)	Path-based Analysis (e.g., network penetration analysis)	Random Sampling with Repeated Trials (e.g., scenario testing)	Hazard Analysis, Risk Prioritisation	Threat Impact Analysis, Risk Scoring
Mathematics	Differential equations (e.g., modelling the dynamic behaviour of threats and defences)	Objective functions (e.g., optimising defence strategies)	Probability distributions (e.g., modelling event likelihoods)	Graph Theory (e.g., network robustness analysis)	Statistical Analysis (e.g., estimating risk exposure)	Risk Formulas, Safety Margins	Threat Impact Analysis, Risk Scoring
Behaviour	Centralised system behaviour (e.g., overall system response to threats)	Decentralised individual behaviour (e.g., interactions between vehicles, hackers)	Sequential processing (e.g., sequence of cyber events)	Network Formation (e.g., communication network vulnerabilities)	Uncertainty Modelling (e.g., predicting the success of cyberattacks)	Identifying and mitigating potential failure points	Categorising and addressing threats based on STRIDE elements
Suitability for System-Level CAV Security	Highly suitable for system-level cybersecurity assessments because it models system-wide feedback loops, accumulations, and information flows.	Less effective for high-level, system-wide CAV security evaluations. Suitable for exploring specific attacker or vehicle interactions.	Primarily deals with discrete events. Effective for specific incidents but doesn't capture broader system dynamics.	Useful for analysing network vulnerabilities but lacks system-wide modelling of the evolving cybersecurity landscape.	Effective for estimating probabilities and risks but doesn't provide a holistic view of system-wide interactions.	Valuable for risk assessment but lacks continuous modelling of dynamic system behaviour.	Focused on identifying and addressing specific threats, not on modelling the entire CAV security ecosystem.

service providers, underscoring the apparent nature of their involvement.

"Systems change" refers to changes in a system's structural, relational, and institutional makeup and its subsystems that address the root causes of a problem. From a system dynamics standpoint, efforts to implement system change must consider how system components are linked together, not just what they are, because component coupling enables the identification of fundamental causes of issues (Hirsch et al., 2007). Moreover, by analysing the connection of system components, one may explain why certain system change attempts result in no visible change while others result in change that is contrary to our desires and intentions (Khan et al., 2021b).

This study introduces an SFM to gauge the impacts of various factors on CAV cybersecurity, illustrated in Fig. 1, which demonstrates the high-level interactions among six key elements of the proposed architecture of CAVs cybersecurity SFM: CAVs Communication Safety, CAV Adopters, CAVs Logfiles, CAVs Hacker's Capability, CAVs Criminology Theory Maturity (CMT), and Public CAVs Cyber Safety Awareness. Each element's detailed architecture description, including the relationships between different components is explained in detail in respective subsections. The model is developed within the system dynamics framework, with key parameters represented as Stocks (levels) and Flows (rates), where Stocks accumulate or deplete based on incoming or outgoing Flows over time.

In the SFM, Stocks can be metaphorically envisioned as "reservoirs" that hold resources or values over time. For instance, in the case of *Technology Readiness and Maturity (TRM)*, this stock accumulates technological advancements over time as new technologies are introduced

and mature. Squares symbolise stocks, and their values are determined by the net effect of the Flows, represented by double-lined arrows. For example, the *Technological Maturation (TM)* flow governs the rate at which *TRM* changes.

Mathematical relationships between Stocks and Flows govern the dynamics of the system. The basic equation governing the relationship between a Stock and its inflows and outflows is presented in Equation (4.1). The equation integrates the difference between inflows and outflows over time alongside the initial value of the stock:

$$Stock(n) = \int_{n0}^n [inflow(n) - outflow(n)] dn + Stock(n0) \quad \text{Equation. 4.1}$$

The values of the stocks evolve based on the accumulation of the inflows and outflows. Positive inflows lead to an increase in the stock's value, while negative inflows (or positive outflows) cause the stock to decrease. This dynamic behaviour is essential for understanding how key parameters such as TRM or CAV Adoption change over time and how they influence CAV cybersecurity outcomes.

In system dynamics modelling, the differentiation between Stocks (representing accumulations or states) and Flows (representing activities or processes that influence the stock values) is critical. For example, in the context of this study, stocks such as "CAV Adoption" or "Public Cyber Safety Awareness" aggregate over time-based on inflows like "Adoption Rate" or "Awareness Campaigns." Meanwhile, exogenous parameters, such as external policies or technological advancements, are treated as fixed or variable inputs that impact the system but are not part of the internal feedback mechanisms.

Additionally, the model includes various dependencies between

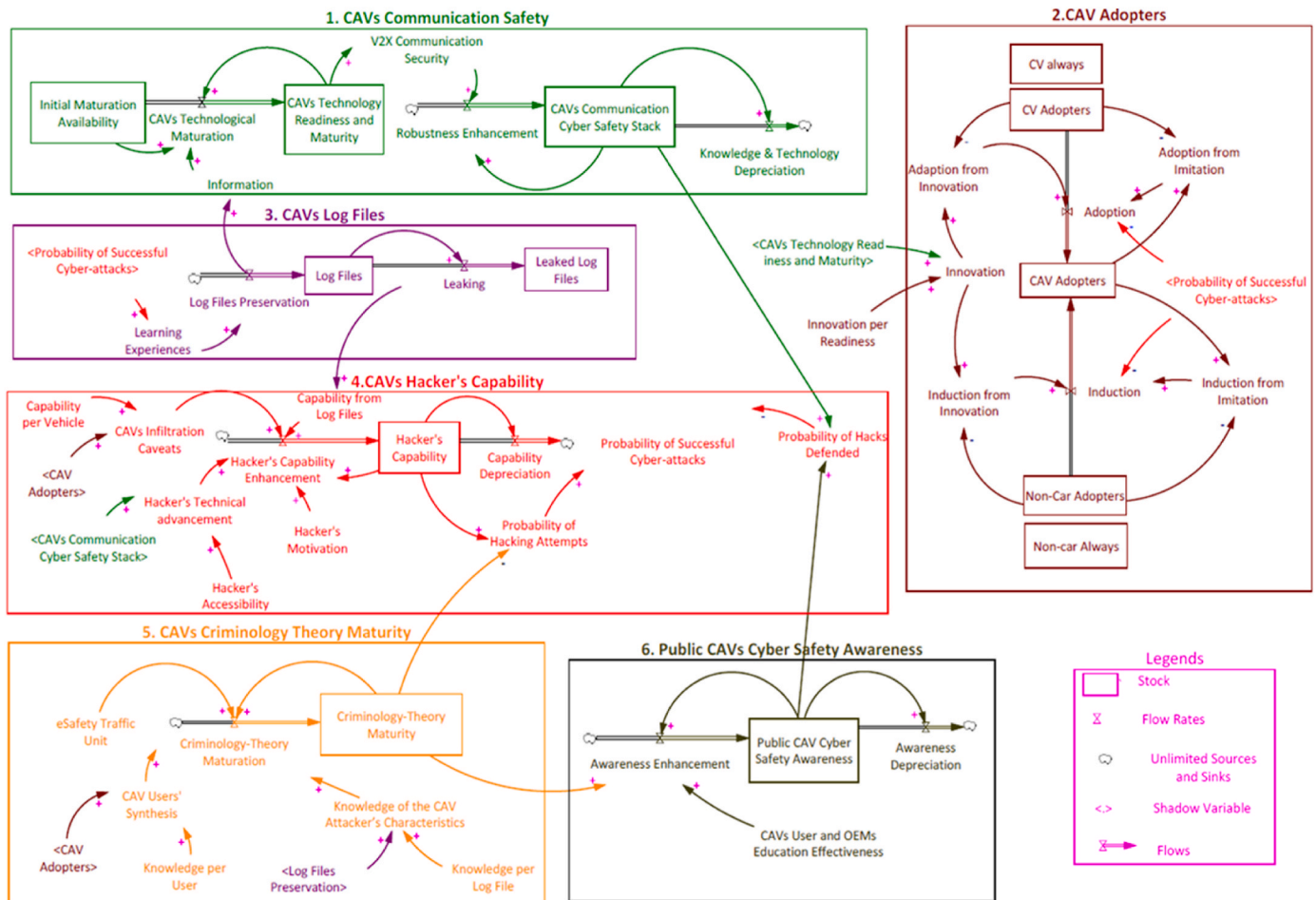


Fig. 1. The proposed architecture of CAVs cybersecurity SFM.

variables, indicated by single-lined arrows, representing either positive (increasing) or negative (decreasing) relationships. The 'tap' symbol in the diagrams signifies flow rates—how fast certain changes occur—while the 'cloud' symbol represents infinite sources or drains, indicating factors beyond the system boundary, such as external resources or global technological trends.

In this specific case, the model captures the interactions between variables like CAV cybersecurity (CAV-CS), log files, hackers' capabilities, CAV adoption, CMT, and public cyber safety awareness. The process through which these variables affect the overall cybersecurity of CAV systems is elaborated in subsequent sections, detailing the specific mechanisms by which flows alter the state of the stocks and how system feedback loops influence long-term behaviour.

4.1. The CAVs communication safety

The primary enabler of cybersecurity is CAVs Communication Safety (CAVs-CS). The mechanism by which CAVs-CS mature is illustrated in Fig. 2. CAVs Technology Readiness and Maturity (TRM) is a fuzzy concept that lacks direct measurement. Nevertheless, it is conceivable to map variables to enable the precise measurement of factors influencing CAVs' technological maturity through empirical observations. Albert (2016)'s operationalised approach, which is subsequently based on the S&D Model (Sommerlatte and Deschamps, 1985), is used to assess TRM. The CAVs Technological Maturation (TM) is the process of continuously de-risking the operation of CAVs in ITS. TM growth is a function of Research and Development, Resources, Processes, and Information (Li et al., 2019).

Businesses and academic institutions make steady progress towards trial versions of emerging technologies by carefully allocating R&D resources to the most exciting projects and taking calculated risks (Cho et al., 2016). The proximity of diverse resources and collaborators within each hub fosters a greater capacity for innovation. The Australian Defence Science and Technology Group identified nine maturity levels for gauging technologies during a program's acquisition phase (Australia- Government, 2020). CAVs Communication Cyber Safety Stock Max enables diminishing returns, i.e., increasing input results in diminishing gains in output. CAVs TRM improves V2X Communication Security, resulting in Robustness Enhancement—the deterrence of cyber-attacks (Khan et al., 2020; Vimmerstedt et al., 2015). The Log File Preservation (including lessons learned) is a useful input to Information. Initial Maturation Availability is the initial non-zero value of the CAVs TRM.

Schumpeter (1939) classical theory identified three components of technological change: i) invention-creation of a new concept, ii) innovation-development of the concept, and iii) diffusion-dissemination of the product, i.e., deployment of V2X Communication Security for Robustness Enhancement in Fig. 2. A more resilient network will improve

CAVs-CS, make it difficult for hackers to infiltrate, and lower the PHA (Tu et al., 2019; Qiu et al., 2019). CAVs-CS will enable a CAVs cyber risk management lifecycle, including automated protection (layering, patching, and backup) and a risk-based approach to resilience. Additionally, it insulates against CAVs physical or proximity access attacks. Furthermore, if institutions that create norms and standards for how to utilise knowledge do not embrace it effectively, knowledge may be disregarded or devalued. (Lundvall, 2010). Appendix A outlines the parameters and expressions of the model.

4.2. CAV adopters

With more CAVs on the road, ITS technology will become mature and widespread. On the other hand, as more CAVs hit the road, threats will increase, hackers experiment with new revenue streams. Stasinopoulos et al. (Stasinopoulos et al., 2020)'s methodology is employed to measure CAV Adopters, as depicted in Fig. 3. Regarding adoption, the Bass Diffusion Mode (BDM) aptly represents CAVs penetration; it permits individuals to adopt CAVs without discarding them (Sterman, 2000). CAV Adopters' assembly depends on non-CV user induction and Conventional Vehicle (CV) approval. The BDM's two crucial parameters are: i) the product's appeal—the innovation coefficient, and ii) product's attractiveness — imitation coefficient. The process is comparable to generating an S-shaped curve, where a sizable segment becomes early adopters as the technology progresses; once the technology matures sufficiently, another subset of the populace persists in embracing the technology. The group of CV Adopters hinders Adoption from Innovation to those embracing CAVs. The CV Adopters are determined by the difference among the CV adopters initial and cumulative shift in adoption.

The impact of Innovation and Imitation on Induction and Adoption could be depicted as a slow or fast scenario, respectively (Shabanpour et al., 2018), while the overall quantity of Car Adopters is determined by summing up the numbers of CAV and CV Adopters. Additionally, Non-car Adopters are the population segment most prone to adopt CAVs due to their accessibility. This group encompasses public transport users, cyclists, children, and seniors (Stasinopoulos et al., 2020). Furthermore, successful cyber-attacks will negatively affect the product's (CAV) Adoption and Imitation, while CAVs-TRM acts as an incentive for Innovation. Appendix A delineates the model parameters and expressions.

4.3. Log files

The main data sources for network visibility in CAV functioning are Log Files. These files document all operations of CAVs executed in ITS, including user behaviour, computing dataflow, and communication signals. Keeping Log Files for a certain amount of time in cloud computing environments can enhance the cybersecurity resilience of CAVs, bolster the effectiveness of ADAS, and offer substantial data

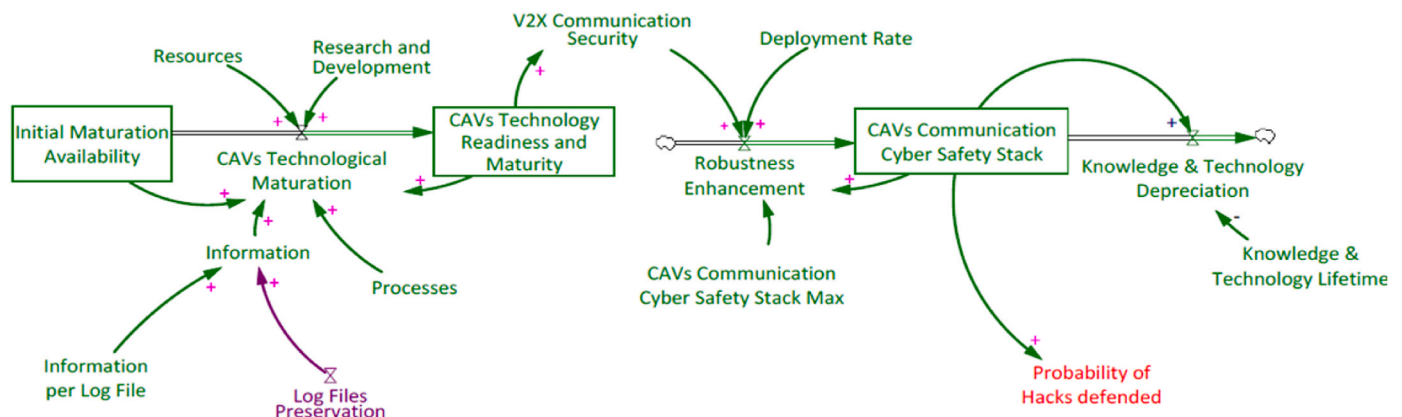


Fig. 2. The CAVs communication safety (CAVs-CS).

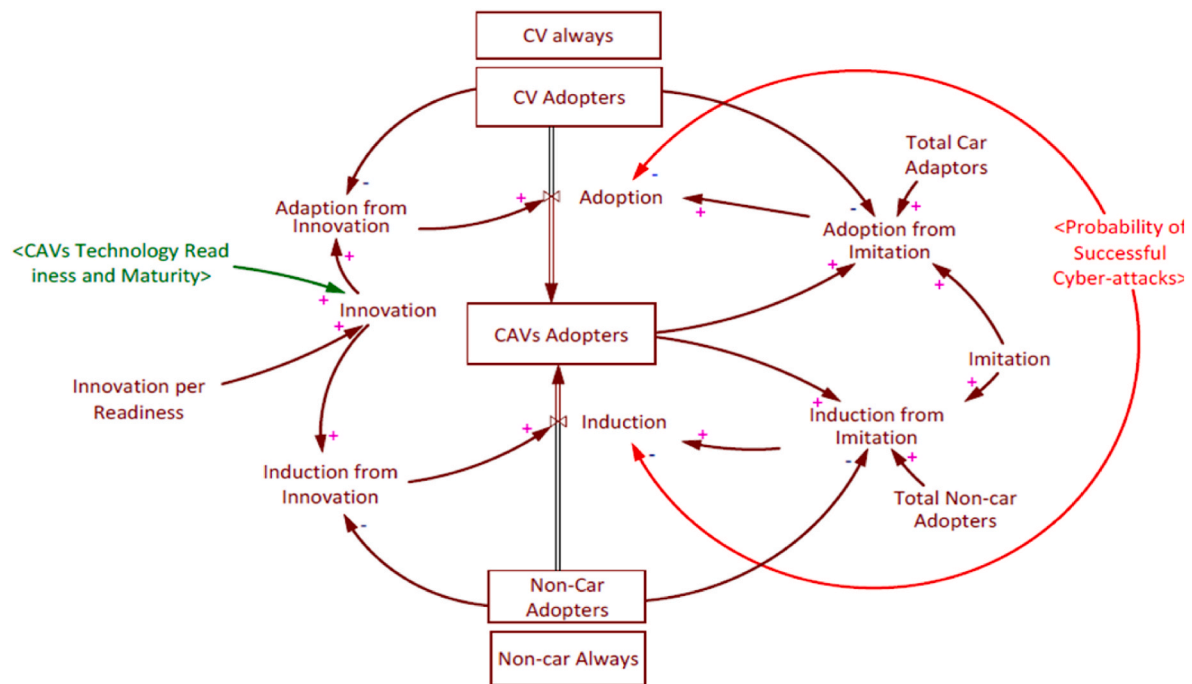


Fig. 3. CAVs adopters.

inputs for informational purposes (Prasad and Rohokale, 2020), as demonstrated in Fig. 4. In the same vein, Log Files are crucial data assemblies, notably in cyber forensic analysis (Kent, 2016). They could aid in reducing harm by supporting evidence for assigning insurance responsibility and advancing future collision prevention tactics (Dimitriadis et al., 2020). In addition, cyberattacks facilitate *Learning Experiences* or the control and essence of hackers’ motivations. A typical example is the reconfiguration of vehicle safety after a crime – implementing car alarms, immobilisers, and central locking to deter vehicle theft. However, the preservation of CAVs Log Files presents a risk—valuable information for cybercriminals, especially when log files are leaked. Appendix A describes the model parameters and expressions.

4.4. The Hacker's capability

The emergence of CAV technology, a mobile CPS, introduces a new domain where cyber assailants, often called "contemporary pirates," would operate. Fig. 5 represents *Hackers' Capability* strategies, encompassing a diverse spectrum of experts leveraging their technical expertise within a digitised environment to accomplish specific objectives. Hackers can be classified into three main types: i) those driven by personal or monetary motives, who breach digital devices and networks, and may engage in activities such as espionage, exhibitions, or simply seeking thrills; ii) those contracted or employed by organisations and

governmental entities to serve as technical experts, assessing vulnerabilities through ethical hacking practices; and iii) hacktivists, who employ hacking techniques to further ideological or political causes (Seebruck, 2015; Zhang et al., 2015). The likelihood of a cyberattack on CAVs is influenced by various factors, including the allure of the target, the motivations of the attackers, the perceived vulnerability of the intellectual assets associated with CAVs, and the preventive measures implemented within the ITS (Aryee, 2020; Choi et al., 2020). The *Hacker's Motivation* plays a significant role in enhancing their capabilities. At the same time, other contributing factors include the *Capability per CAV Log File*, the *Technical Advancements* of hackers, and CAVs *Infiltration Caveats*.

As the count of CAV adopters increases on the road, criminals relentlessly search for new sources of income, making it easier for hackers to discover *Infiltration Caveats* (Huq et al., 2017; Ghadi et al., 2020). The *Hacker's Technical Advancement* is an inadvertent aftermath of CAV-CS, constrained by the *Hacker's Accessibility*, as shown in Fig. 5. Similarly, the *Leaked* log file enables hackers to conduct cyberattacks. *Hacker's Capability Max* permits diminishing returns; as the maximum value is reached, the improvement will be minimal at a high cost. *Capability Depreciation* refers to the gradual decline in the *Hacker's Capability* over time (Lundvall, 2010), limited by the *Capability Lifetime*. Furthermore, an increase in the *Hacker's Capability* causes an increase in the *PHA*, which CMT may constrain. Appendix A describes the model

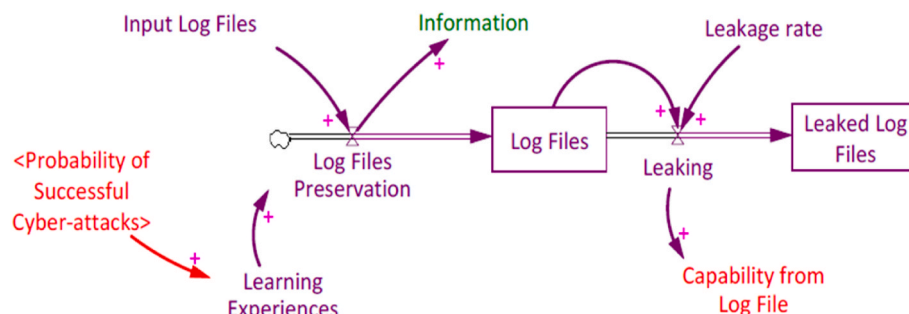


Fig. 4. Log files (CAVs).

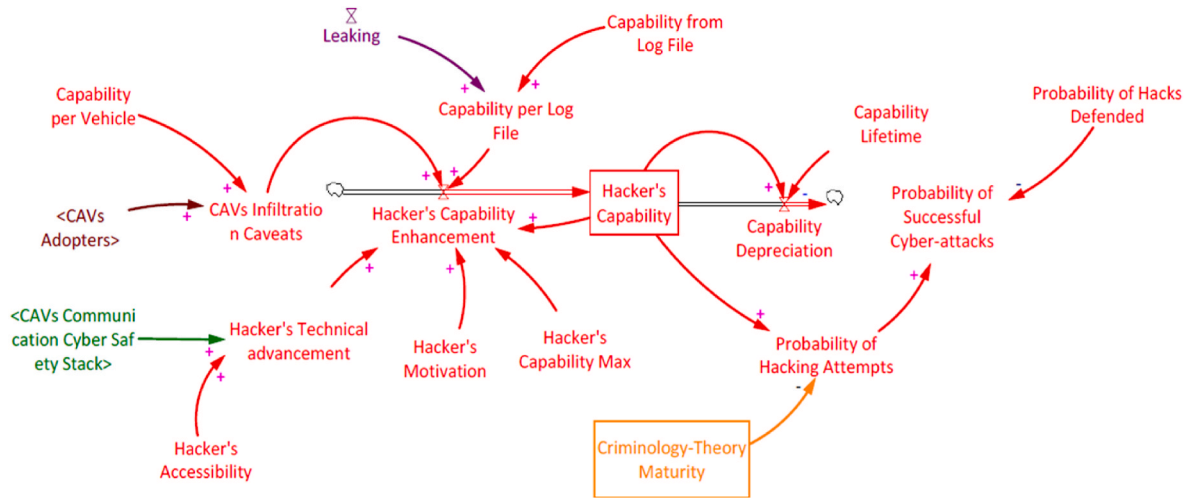


Fig. 5. The Hacker's capability.

expressions and parameters.

4.5. CAVs criminology theory maturity

Indeed, crime design is often an afterthought for new technology, and CAV is no different. The pervasiveness of CAVs connectivity expands the scope of crimes committed in both cyberspace and physical space in ITS. Various theories illustrating the rapidly changing, dynamic, and expanding environment for transportation-related criminal possibilities are: technology life cycles, the social construction of technology, market replacement approaches, economic path dependency, and the most straightforward one—Routine Activity Theory (RAT) (Newton, 2017). The use of RAT is a powerful technique for reducing hacker capabilities and the cyber risks posed by CAVs (Kennedy et al., 2019).

The *Criminology Theory Maturation* is contingent on the *eSafety Traffic Unit* (eSTU), *CAV Users' Synthesis*, and *Knowledge of CAV Attackers' Characteristics*, as shown in Fig. 6. The CAV adopter facilitates mounting up the essential knowledge for synthesising *Users' Synthesis*, utilising their existing expertise and evolving knowledge of CAV operations to counter criminal activity in the ITS domain through the eSTU. Knowledge of CAV cyberattackers' characteristics (intentions or behaviours), i. e., poor socioeconomic position, hyperactivity, dark triad personality traits, and socialisation towards law-breaking behaviour reduces the probability of an attack (King et al., 2018; Holt et al., 2010). Furthermore, CAVs CMT provides a good baseline for *Awareness Enhancement* in *Public CAV Cyber Safety Awareness*. Appendix A describes the model

expressions and parameters.

4.6. Public CAVs cyber safety awareness

Road safety now encompasses a new area of concern: the cybersecurity of CAVs. In this context, human vulnerability is the most probable cause of a successful cyberattack. To address this issue, researchers specialising in human factors and psychology can contribute to improving CAV cybersecurity by exploring ways to minimise the risk of successful attacks. This exploration would incorporate a human-centred approach considering the adversary's ability to choose the type of assault. According to Trend Micro, a staggering 91 per cent of successful cyberattacks originate from the weakest link in the system—the human element (TrendMicro, 2020). Fig. 7 illustrates the mechanism for *Public CAVs Cyber Safety Awareness*, visually representing the process. The two main facilitators of *Awareness Enhancement* are *CAV Users and the OEM's Education Effectiveness*, and *CMT*.

The dynamic improvement of CMT would provide the baseline features of CAVs cyber-attackers (Holt et al., 2010; Wilson and Hash, 2003). Education is critical for properly teaching and protecting CAVs and their customers against cyberattacks. Individuals differ in their capacity to appropriately recognise cybersecurity hazards; for example, 23 per cent of people manage less than half of cybersecurity circumstances, while just 4 per cent handle more than 90 per cent of situations (Katerina and Nicolaos, 2018). The Harvard Business Review report describes human error as a primary driver behind the majority of cyber incidents

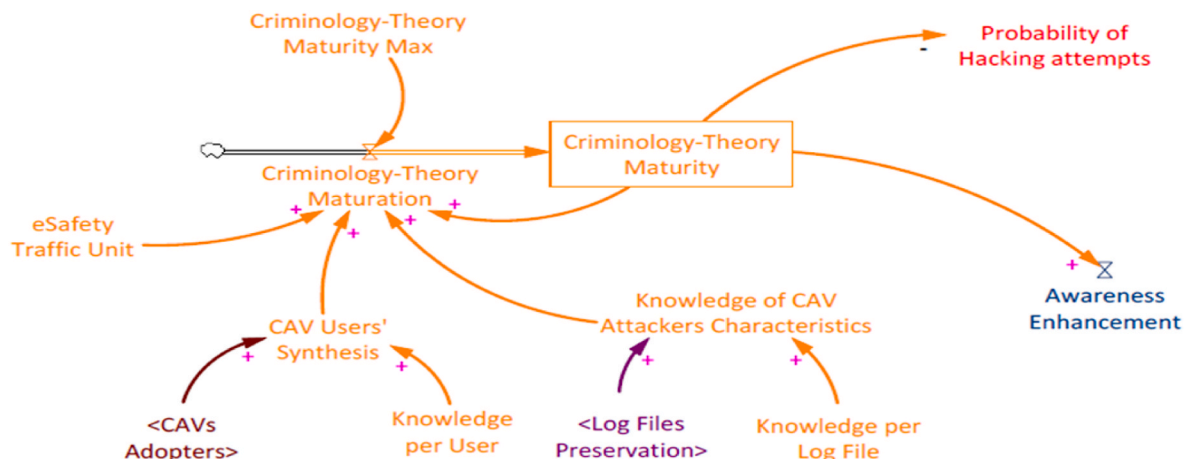


Fig. 6. CAVs criminology theory maturity.

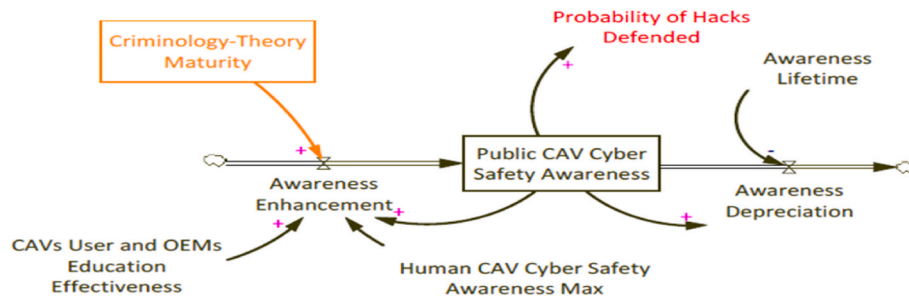


Fig. 7. Public CAVs cyber-safety awareness.

(Chamorro-Premuzic, 2023). The absence of accurate information and awareness among key stakeholders and the general public is a fundamental barrier to successful risk management (Cohen et al., 2007; Christiansen and Piekarsz, 2018). The *Awareness Depreciation* constrained by *Awareness Lifetime* could deplete *Public CAV Cyber Safety* if there is a lack of adequate guidelines for its use (Lundvall, 2010). Appendix A describes the model expressions and parameters.

5. Model testing

To ensure the reliability and validity of the proposed SFM, we conducted a rigorous evaluation process following the established *guidelines in the literature*. This evaluation included both *structural and behavioural testing* (Stermann, 2000; Stasinopoulos et al., 2020; Valencia Arias and Obando Montoya, 2012)(Stermann, 2000; Stasinopoulos et al., 2020; Valencia Arias and Obando Montoya, 2012).

5.1. Structural validation

We assessed the SFM's parameter assessment, boundary adequacy, dimensional consistency, extreme conditions, and overall structure during structural testing. Key findings from this phase included.

- **Boundary Adequacy:** The model demonstrated high sensitivity to removing existing endogenous structures while introducing new endogenous structures, which had a relatively minor impact.
- **Dimensional Consistency:** It is checked that all expressions in the model are dimensionally consistent, ensuring that units on both sides of each expressions match.
- **Parameter Evaluation:** We confirmed the conceptual alignment of all model parameters with those of the existing system and ensured that the units on both sides of each expressions were consistent. For instance, during parameter evaluation, we scrutinised the conceptual alignment of all parameters with those of the existing system, such as adjusting the *Hacker's Motivation* to ensure that it positively impacts the *Hacker's Capability*.

The results of these tests provided valuable insights for model refinement and subsequent recalibration.

5.2. Behavioural validation

Behavioural testing involved evaluating the SFM's performance under extreme conditions, reproducing known behaviours, assessing boundary adequacy, and identifying potential anomalies.

- **Extreme Conditions:** By systematically altering individual parameters (e.g., V2X communication security), we verified that the model's output (e.g., PHD and PHA) remained within expected ranges, indicating realistic behaviour.
- **Behaviour Reproduction:** Due to the lack of historical CAV cybersecurity data, we compared the model's output with projected

cybersecurity trends for CAVs (Khan et al., 2021a) to assess its ability to replicate expected behaviours.

- **Policy Testing:** We also evaluated the model's response to various policy interventions that can help assess its usefulness for decision-making (detailed in the next section)

Overall, the rigorous testing process has strengthened our confidence in the SFM's ability to accurately represent and predict the dynamics of CAV cybersecurity.

6. Simulating policy impacts

After model development and testing, the next phase involves investigating plausible scenarios to determine effective CAV cybersecurity policies. SD scenario simulation is conducted in the literature by incorporating various data sources, such as existing empirical and theoretical literature, the combination of quantitative or qualitative data, secondary data analysis, and expert opinion. This comprehensive approach enables researchers to explore strategies for system change. For example, Nazareth and Choi (2015) used a normalised scale (0–1) for various input parameters for information security in addition to available data; Lounsbury (2002) utilised quantitative data and experts' opinions to model epidemic changes; and Repenning (2002) developed an SD model entirely from descriptions of parameters in the literature to account for companies' failures and successes with innovations. Therefore, the model's scenario simulation data is sourced from two methods: utilising quantitative data for variables with measurable values and employing exploratory data to quantify hidden variables to reproduce the observed behavioural patterns of actors within the system (Hirsch et al., 2007). Table 3 illustrates the combinations of scenarios simulated.

SD model runs are analysed with enhanced intuition, provide a dynamic interaction of variables affecting CAVs' cybersecurity, and highlight interventions that could have unintended consequences when CAVs are deployed in ITS. The aim is to achieve a delicate equilibrium between promoting CAVs adoption and establishing a resilient cybersecurity framework while systematically envisioning the potential paths of CAVs deployment, understanding the impact of policies and technological advancements on CAVs cybersecurity, and highlighting the interconnected dynamics.

For scenarios simulation, the total *Population* is scaled to 1, allowing each component to effectively demonstrate the relative impact of CAVs by representing a proportion of the total population (Stasinopoulos et al., 2020). Similarly, the total count of *Car Adopters* is determined by summing up the numbers of CAV and CV *Adopters*. Additionally, *Non-car Adopters* comprise the population segment most likely to adopt CAVs due to their accessibility. This group encompasses public transport users, cyclists, children, and seniors (Stasinopoulos et al., 2020). CV *Adopters* are measured as the difference between the CV *adopters initial* and cumulative changes in adoption. Based on data from the Australian Statistics Bureau, it is estimated that the total value of CV *always and CV Adopters initial* constitutes approximately 57% of the overall ownership of passenger vehicles.

Table 3
Combinations of scenarios simulated.

Scenarios	CAVs technology readiness and maturity	Criminology Theory Maturity	CAV users and OEMs education	CAVs penetration	CAVs penetration with public behaviour analysis	Potential Impact
Baseline	The CAVs <i>communication security</i> is reinforced over time as the technology matures.	Not included	Moderate	Moderate (Only by adoption)	Moderate	CAVs-CS is improved, and PHA increased as well as PHD (as expected).
Criminology Theory Maturity	The CAVs <i>communication security</i> is reinforced over time as the technology matures.	Utilising the criminology theory, i.e., the use of RAT is a powerful technique for reducing hacker capabilities and cyber-risks posed by CAVs.	Moderate	Moderate (Only by adoption)	Moderate	Significant decrease in PHA. There is a slight increase in CAV Adopters when the Probability of Successful Cyber-attacks decreases.
CAV users and OEMs education	The CAVs <i>communication security</i> is reinforced over time as the technology matures.	Not included	Robust CAVs user and OEMs education.	Moderate (Only by adoption)	Moderate	PHD is augmented through a reinforcing mechanism that runs through CAV Users and OEMs Education Effectiveness. A rise in CAV Adopters
CAVs penetration	The CAVs <i>communication security</i> is reinforced over time as the technology matures.	Not included	Moderate	Adoption + induction	Moderate	
CAVs penetration with a robust human behaviour analysis	The CAVs <i>communication security</i> is reinforced over time as the technology matures.	Not included	Moderate	Adoption + induction	Robust public CAVs behaviour synthesis	A decrease in PHA and an decrease in PHD.

The values for *Innovation* and *Imitation* in the speedy scenario are sourced from (Shabanpour et al., 2018), while the values for the slow scenario are obtained from the study conducted by (Lavasan et al., 2016). *Non-car Adopters* initial value in the context of CAVs penetration is set based on the literature range of 0.02–0.6, as described by (Harper et al., 2016; Litman, 2020), aiming to achieve CAV induction within that range, with the market size for CAVs projected to be at least 40% (Shabanpour et al., 2018; Menon, 2015). The exogenous input variables are scaled on a normalised range of 0–1, as outlined in Table 3, with the values selected in the middle to reflect a moderate scenario, as suggested in the literature (Nazareth and Choi, 2015).

The system dynamics are simulated utilising Vensim® Software, Version 8, employing the Runge-Kutta integration method. The scenario simulation duration is 50 years, considered sufficient to capture the hype cycle of CAV technology (Bartl, 2015). This time frame covers the entire spectrum, from the initial conception of CAVs to their widespread adoption by a significant portion of the population. For example, passenger car manufacturers initiated Anti-lock Braking Systems (ABS)

trials in the 1960s. However, it took approximately four decades for ABS to achieve widespread prevalence, with its adoption becoming prominent during the 1990s.

The **Baseline (Technological driven only)** scenario assumes that CAVs-CS is reinforced over time as the technology matures. The criminology philosophy for countering CAV cyber-crimes is not included. *CAV Users and OEMs Education Effectiveness*, and *Public CAVs Cyber Safety Awareness* are moderate. The composition of CAV Adopters is from *Adoption* only. The change in CAVs-CS is amplified because of a reinforcing mechanism that runs through the CAVs TRM. As a result, CAVs-CS is improved, and PHA increased as well as PHD (as expected), depicted in Fig. 8. The CMT scenario utilises theory enabled by CAV Users' *Synthesis*, the *eSTU*, and *Knowledge of CAV Attackers' Characteristics*. The change in PHA is resisted because of a balancing mechanism that runs through CMT. As shown in Fig. 9, this action results in a significant decrease in PHA. There is a slight increase in CAV Adopters when the *Probability of Successful Cyber-attacks* decreases.

The **CAV users and OEMs education** scenario marks an

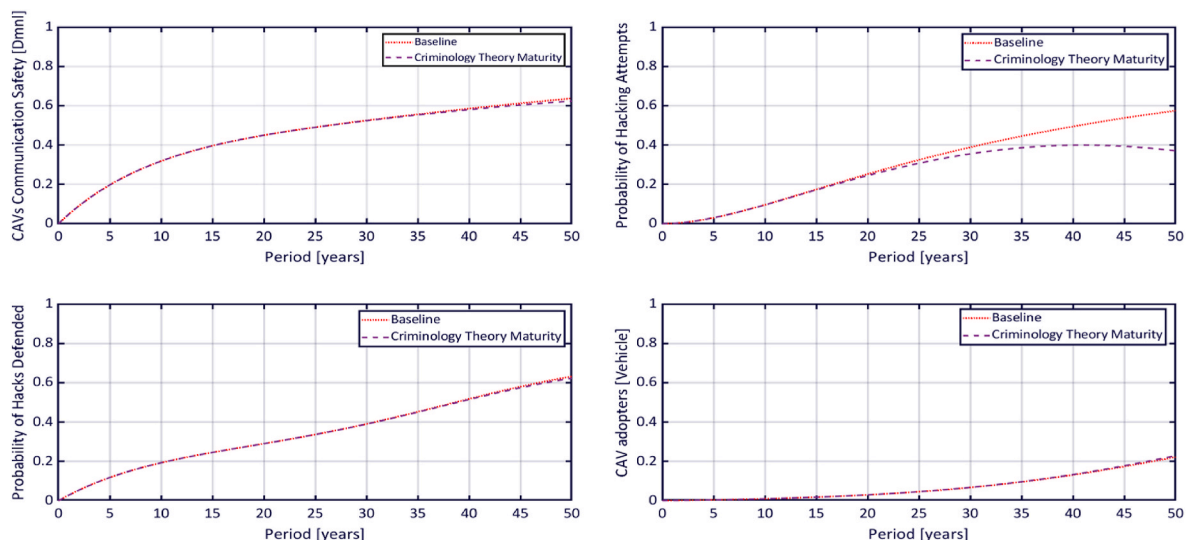


Fig. 8. Cyber-safety assessment: Baseline vs. Criminology theory maturity.

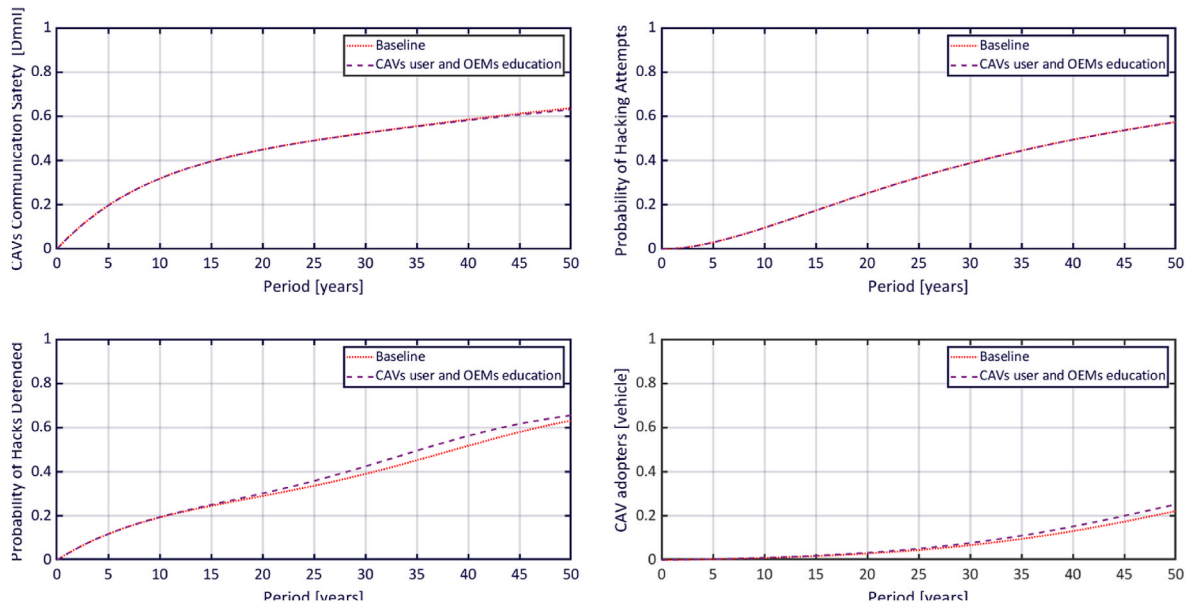


Fig. 9. Cyber-safety assessment: Baseline vs. CAVs user and OEMs education.

improvement in *PHD* only, illustrated in Fig. 10. *PHD* is augmented through a reinforcing mechanism that runs through *CAV Users and OEMs Education Effectiveness*. *PHD* improvements reduce effective cyber assaults, allowing for a rise in *CAV Adopters*. **CAVs penetration** occurs via *Adoption* and *Induction*, leading to a rise in *CAV Adopters*. The change in *PHA* is amplified because of a reinforcing mechanism, i.e., more *CAV Infiltration Caveats*, and a corresponding rise in *PHD*. Nonetheless, in **CAVs penetration with public behaviour analysis**, the change in *PHA* is resisted because of a balancing mechanism, i.e., robust public CAVs behaviour synthesis. As a result, there is a decrease in *PHA* and an improvement in *PHD*.

Similarly, Fig. 11 illustrates the synthesis of the preceding scenarios using a boxplot. The box plot illustrates the symmetry and skewness of the behaviour of the four performance metrics under each scenario. *CAVs-CS* exhibit nearly identical behaviour in all scenarios. The *PHA* decreased in 'CMT' but increased in 'CAVs penetration', as the number of access points grew, making them more susceptible to cyber-attacks.

PHD is enhanced in 'CAV users and OEMs education'. Moreover, in 'CAVs penetration', the *PHD* elevated due to a similar rise in *PHA*; but, 'with human behaviour analysis', the *PHA* showed a noticeable decrease. Due to decreased successful cyberattacks, *CAV adopters* have increased in the 'CMT' and 'CAV users and OEMs education' scenarios.

7. Discussion

The article initiates a discussion on the quantification of cybersecurity in CAVs. This quantitative analysis represents a novel approach compared to the predominantly qualitative methods employed in previous studies. By utilising the SD approach, we have constructed a comprehensive model that captures the essential parameters influencing the cybersecurity of CAVs within the ITS. Through systematic simulation, the established model effectively aligns with the primary objective of developing a robust framework for CAVs cybersecurity. Additionally, the secondary objective is to illustrate the dynamic nature of

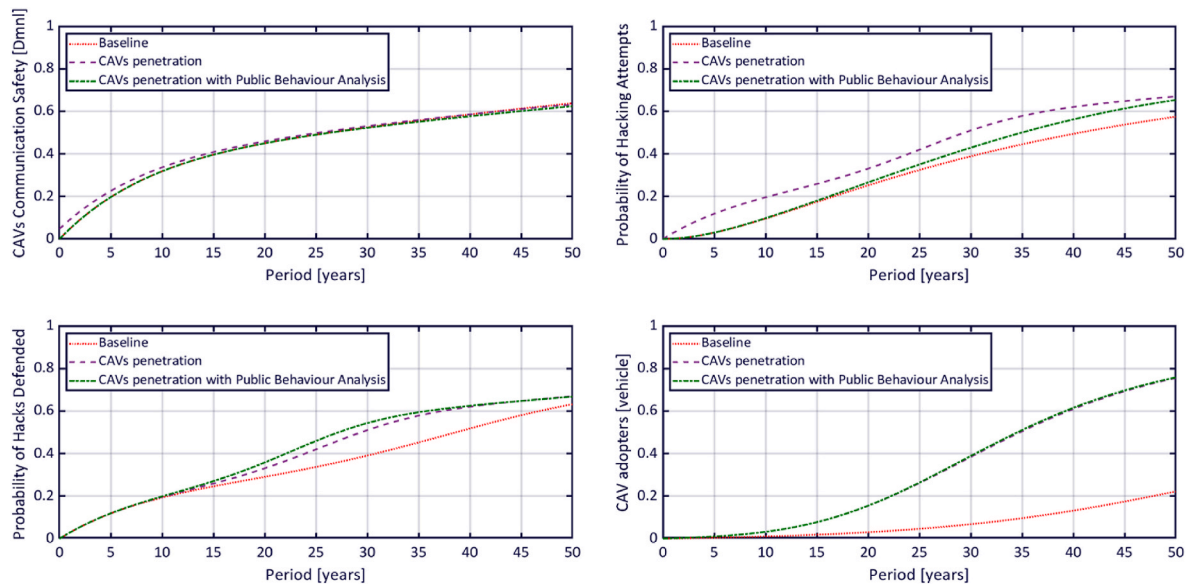


Fig. 10. Cyber-safety assessment: Baseline vs. CAVs penetration.

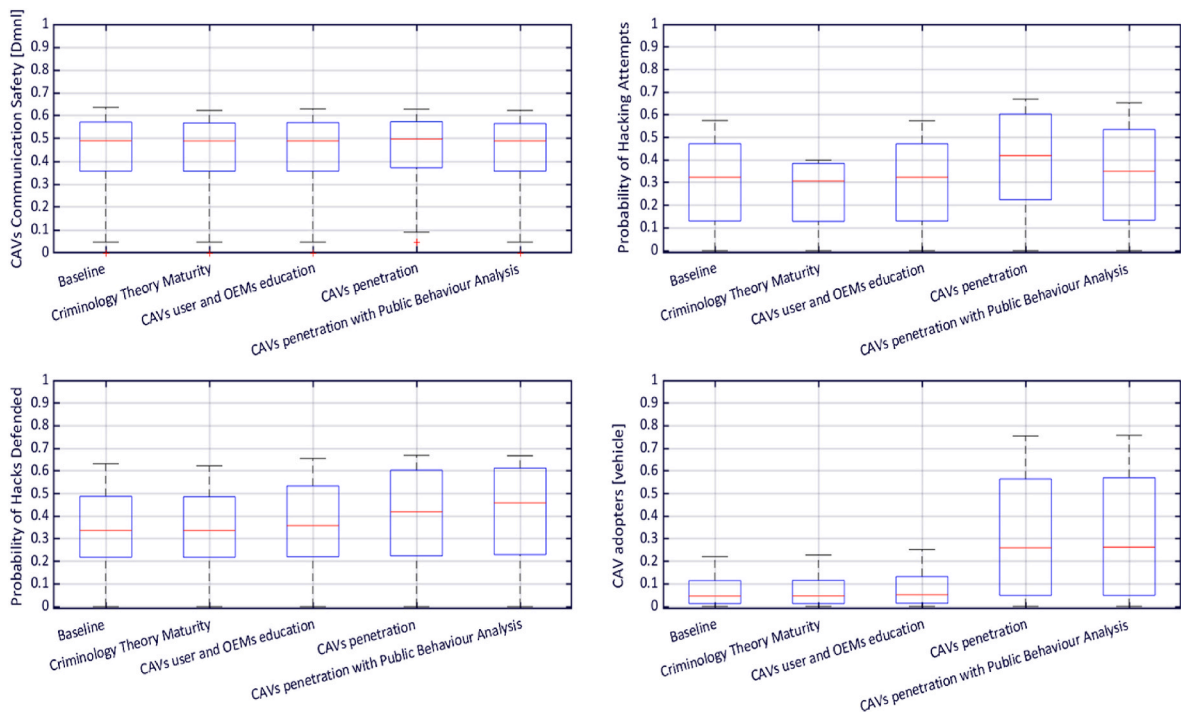


Fig. 11. Box plot visualisation of scenarios simulated for CAVs cybersecurity assessment.

interactions among variables impacting CAV’s cybersecurity, shedding light on interventions that may inadvertently yield unintended consequences when CAVs are integrated into the ITS.

In the context of real-world CAV deployment, these findings have critical implications for cybersecurity strategies and risk management. The widespread deployment of CAVs within ITS introduces complex

cybersecurity challenges that must be addressed holistically. Policy resistance refers to the system’s (ITS) reaction to any intervention (e.g., CAV deployment), which can negate the effectiveness of such interventions if not properly managed (Stermann, 2000). The key stakeholders in CAV deployment include OEMs (original equipment manufacturers), technology partners, communication and cloud service

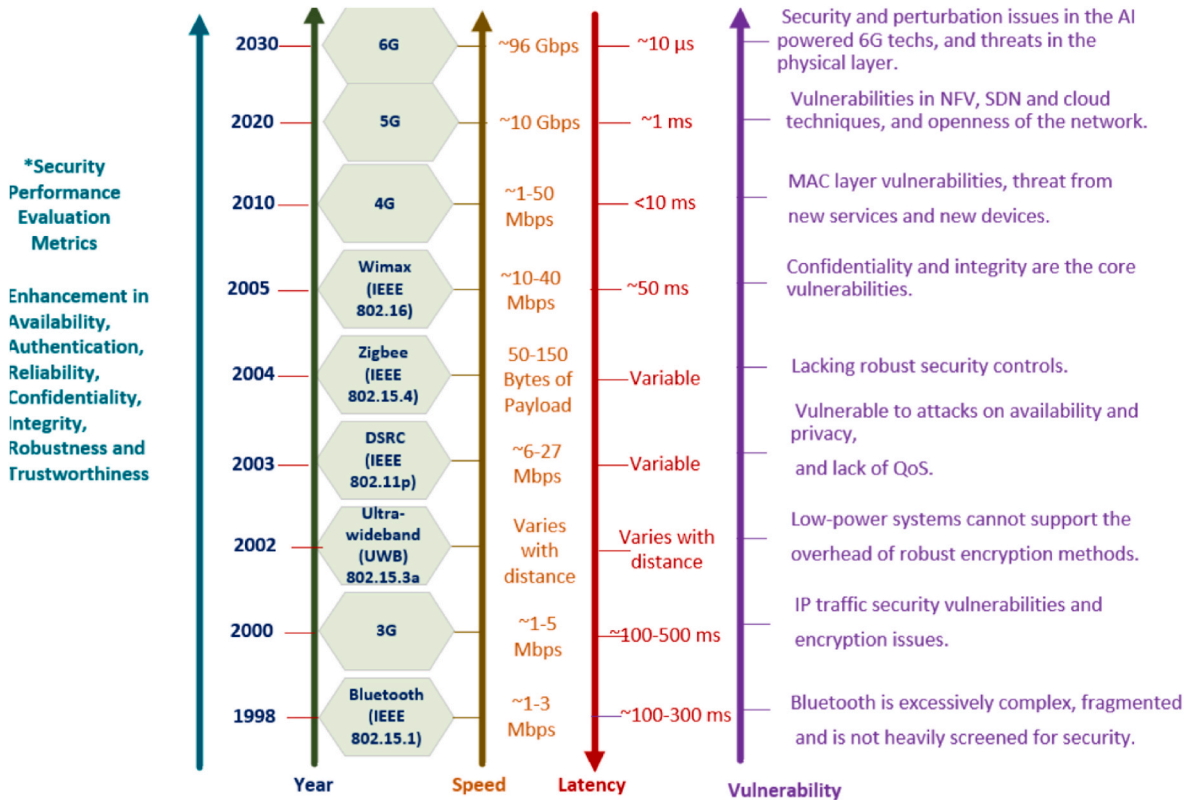


Fig. 12. V2X communication (Source: The authors’ synthesis).

providers, regulators, policymakers, road and infrastructure authorities, and end-users.

By analysing feedback loops and system interdependencies through the SD model, this study captures the broader cybersecurity challenges of CAVs in a way that isolated studies often miss. SFM findings show that interdependencies between different system components can produce non-linear behaviours, making it difficult to predict outcomes when addressing a single avenue in isolation. This integrated approach provides a more realistic understanding of how multiple variables affect CAV cybersecurity across the entire ITS architecture.

The CAVs-CS is designed to act as the first line of defence against cyberattacks. CAVs-CS helps develop a CAV's cyber risk management lifecycle, encompassing several protective measures, including automated protection mechanisms (such as multi-layer security, patch management, system backups, and proximity access security). As CAV technology matures, the need for a robust CAV cyber risk appetite and a risk-based approach to resilience becomes evident. The system's security maturity can be measured through processes involving Research and Development (R&D), operational resources, and information-sharing protocols (Rosenfeld et al., 2007).

The CAV technology Maturation is the de-risking of CAV operations in ITS. However, innovations also increase vulnerabilities. Given the inherent uncertainty and technological risk associated with CAV technology, assessing the potential technical risks associated with CAV adoption is critical. This can be demonstrated in Fig. 12: Quality of Service (QoS) vs. security provisions for CAVs wireless technologies in terms of speed, latency, vulnerabilities, and security performance evaluation metrics. For example, the security and privacy challenges evident after the convergence of the 5G and 6G networks into V2X (Javed et al., 2022) would include a plethora of cyberattacks, necessitating a proactive security assessment. Similarly, customised designs for ADAS may aid in lowering the overall risk (Chen et al., 2021). Similarly, the dynamic optimisation techniques based on real-time data need further attention.

The ongoing cybersecurity battle between attackers and defenders in the context of real-world CAV deployment is heightened by the attractiveness of CAVs as high-value targets. The likelihood of a cyberattack is influenced by the hacker's perceived opportunities, including Hacker's Capability, the existing deterrent measures within the ITS, and the system's perceived vulnerabilities (Aryee, 2020). In particular, the Hacker's Capability is linked to the Technology Readiness and Maturity (TRM) of CAV systems, the hacker's motivation, the level of CAV penetration in the market, and the effectiveness of log file preservation. By incorporating criminology theories like Routine RAT, this study seeks to understand hacker motivations and profiles, which can help develop more effective countermeasures.

The maturation of criminology philosophy plays a significant role in advancing the understanding of CAV cybercrime and criminal justice (Kennedy et al., 2019). For example, incorporating RAT into cybersecurity strategies can help reduce Potential Hacker Attacks (PHA), as depicted in Fig. 8. A decrease in successful cyberattacks can increase market confidence, making the CAV ecosystem more attractive for CAV Adopters. However, for criminology theory to keep pace with evolving cyber threats, specialised, i.e., eSTUs, will need to be developed. These units would focus on understanding the evolving capabilities of CAV hackers and designing strategies to mitigate their impact.

The study also identifies CAV Adopters as a crucial factor in ensuring widespread acceptance of the technology. As CAVs are deployed, big data generated by their operations offers critical insights into improving the security of ITS. Therefore, the safety of CAV-based ITS may be assessed using data models and applications (Lian et al., 2020). On the flip side, as the quantity of CAVs on the road increases, hackers can explore novel models for generating profit. The growing number of access points in CAVs raises the PHA (shown in Fig. 10), increasing cyberattacks. Increased cyberattacks will undoubtedly derail CAV adoption, necessitating an adaptive, holistic approach to avert this

outcome.

Nonetheless, cybersecurity for CAVs requires upskilling the operation's weakest link—the human being (Annarelli et al., 2020). According to the Ponemon Institute's 2019 Report, employee carelessness is the underlying cause of 24 per cent of data breaches, which is significant (IBM, 2019). Queensland Business Technologies reports human error infections of 95% in 2017/2018 (QueenslandBusinessTechnologies, 2019), and human error accounts for 90% of data breaches, according to TechRadar's findings (Spadafora, 2019). Introducing a CAV-operated license and awareness-raising programs tailored to specific stakeholder groups will benefit CAV cybersecurity (illustrated in Fig. 9).

Similarly, proper information dissemination management, test-ride events and transparency on safety solutions would all have an impact on CAV's overall social acceptability, improving cybersecurity. The scenario 'CAVs penetration with public behaviour analysis' has a noticeable decrease in PHA (illustrated in Fig. 10). Characterising individuals vulnerable to cybersecurity failures, managing the impact of reduced driver-driving abilities, and understanding the motives and attributes of CAV cyberattacks (psychological profile) (Kennedy et al., 2019; Wilson and Hash, 2003; Christiansen and Piekarz, 2018) have the potential to improve CAV cybersecurity.

8. Policy recommendations

This study provides a comprehensive analysis of cybersecurity in CAVs through a SD approach, offering a quantitative framework for understanding and mitigating cyber risks in ITS. The findings emphasise the importance of addressing policy resistance, managing the technological vulnerabilities inherent in CAV deployment, and understanding hacker motivations through criminology theories.

The CAV Penetration scenario highlights the increased risk posed by growing access points in CAV systems, which elevates the likelihood of cyberattacks. Policymakers should implement proactive, multi-layered cybersecurity frameworks that address these risks early. This involves formal regulations, technical standards, and fostering cybersecurity awareness across industries to mitigate potential threats before CAVs become widespread.

CAV operational logs are also a valuable resource for technological maturation, yet they present a lucrative target for hackers. Formulating a systematic policy for CAVs log file preservation is essential, i.e., which logs should be retained? How long will it last? Who should have access to the data? The biggest difficulty lies in striking a balance between three key areas: Privacy and Freedom: How much personal data can be collected from CAV users without infringing on their privacy and freedom of movement? Business Needs: How much access should CAV manufacturers and service providers have to operational data and user information while protecting their investments and innovations? Regulatory Control: How much control should government regulators have over CAVs, ensuring safety without stifling development and innovation?

The CTM scenario shows how advancements in criminological theory can inform the understanding of cybercrime related to CAVs and guide criminal justice responses. Regulators should adopt a systematic approach to cybersecurity oversight, requiring regular vulnerability assessments by manufacturers (OEMs) and ensuring continuous updates to cybersecurity standards. Criminological insights can enhance policy strategies to anticipate and reduce hacking attempts.

The CAV Users' and OEMs' Education scenarios emphasise that public awareness and effective communication play crucial roles in increasing social acceptance of CAVs and strengthening cybersecurity. Stakeholders should invest in public education programs, such as test-ride events and transparent safety communications, to build trust and ensure the public adopts safe behaviours. This will improve overall cybersecurity resilience as public perception aligns with security measures.

Across various scenarios, the study identifies that addressing CAV

cybersecurity requires considering technological, human, and social elements, which involve multiple stakeholders. OEMs, regulators, cybersecurity experts, and criminologists should collaborate to tackle the complex, evolving nature of CAV cybersecurity threats. Coordinated knowledge-sharing and joint risk assessments are essential for a comprehensive response to these challenges.

The CAV-CS scenario highlights technological challenges in securing communication channels within CAV systems, alongside the need for improved log file management and hacker deterrence mechanisms. Vendors and regulators should promote continuous innovation in cybersecurity technologies, such as enhancing communication safety, improving log management, and addressing hacker capabilities (Khan et al., 2024d). This ensures that CAV systems remain secure as new vulnerabilities and technologies emerge.

Given the lack of extensive empirical data, the study shows that SD modelling offers valuable insights for evaluating CAV cybersecurity despite the data limitations. Policymakers should consider leveraging SD models to simulate potential cybersecurity threats and scenarios, aiding in the development of effective, data-informed policies that anticipate risks and guide security strategies.

9. Future direction and limitations

While this study has made a noteworthy contribution to assessing the factors that influence CAVs' cybersecurity, it is important to recognise the constraints of the proposed model. One drawback of SD modelling is the prerequisite of having initial values for all stock variables in the model before commencing simulations. For Flows that are directly proportional to their stocks, initiating with a value of zero is impractical as it would hinder any variation in the flow rate. Hence, a non-zero initial value is presumed to enable meaningful adjustments in the flow. Additionally, some relationships in the model describe phenomena that are difficult to relate directly to the real world. For instance, quantification of CAVs *Technology Maturation* through parameters like *processes*, *resources*, etc. Therefore, these variables are depicted on a normalised scale ranging from 0 to 1, with values chosen towards the centre to portray a moderate scenario.

SFM can manage greater complexity than conventional social science modelling methodologies, but it cannot capture the whole complexity of a process; Modelers must eventually decide which system parts and interactions are relevant to an issue and can be handled in a single model. For instance, factors like regulatory law and trust are not taken into account, as they fall outside the scope of this paper. The choice is primarily driven by the desire for a defined scope and the ability to concentrate on specific parameters through focused analysis. Moreover, the research being undertaken by leading automotive firms such as Waymo, Tesla, Mercedes Benz, Baidu, Uber, Google, and Audi (He et al., 2020; Tesla, 2018), as well as the numerous pilot programs around the world, such as 'Advanced Connected ACV2 trials,' 'Austroads Future Vehicles & Technology Initiative' in Australia (Vicroads, 2021; Austroads, 2021), would aid in the requisite data set for more robust quantification of CAVs cybersecurity parameters. Nonetheless, the scarcity of required empirical data, the subjective nature of cybersecurity, and the high degree of uncertainty make the semi-quantitative approach the preferred method for the foreseeable future.

10. Conclusion

Cybersecurity is a process; system designers need to keep abreast of the developments in cyber-attacks on the CAV-embedded framework while ascertaining the adoption of autonomous driving. A robust cybersecurity framework is comprised of three interconnected links: formal, technical, and cultural. Vulnerability assessment and prior adaptation of CAV's cyber-safety knowledge will boost risk assessments by OEMs and regulators. There is currently less empirical data available on CAVs' cyber-safety due to the evolving nature of CAV technology.

The wait-and-see approach will have significant implications, which could lead to the introduction of corrective steps by the time adequate CAVs are in operation. The system-oriented approach is suitable because of the highly complicated and uncertain nature of CAVs-based ITS, with many components and stakeholders (Kim et al., 2022) and numerous interactions among them, frequently involving multiple feedback mechanisms. This study pioneers the use of SD to facilitate a more nuanced understanding of the ITS system and the relevant parameters affecting CAV cybersecurity. Since SD modelling is deterministic (rather than stochastic), it may aid in useful future insights despite the lack of empirical data in the cybersecurity paradigm of CAVs.

The proposed SD model aims to help assess cybersecurity opportunities in deploying CAVs within the ITS and to help foster the development of effective policy instruments by envisioning how components interact. The SFM is developed by integrating the main elements that lie at the heart of CAVs cybersecurity and shape the model's foundation, namely: *CAVs Communication Safety (CAVs-CS)*, *CAV Adopters*, *Hacker's Capability*, *Log Files*, *Criminology Theory Maturity (CMT)*, and *Public CAVs Cyber Safety Awareness*. The established SFM has its structure and behaviour tested and is used to conduct scenario analyses for CAV cybersecurity using various plausible scenarios.

The findings from various scenarios reinforce the study's conclusions. For instance, the Criminology Theory Maturity scenario illustrates how advances in criminological theory can enhance the understanding of CAV-related cybercrime and inform criminal justice strategies, ultimately reducing the likelihood of hacking attempts. Such insights foster greater market confidence in CAV adoption. In contrast, the CAV Penetration scenario reveals the heightened risks posed by increasing access points, highlighting the need for a proactive, multi-layered cybersecurity approach to mitigate threats and sustain public trust in CAV technology.

Furthermore, the study underscores the importance of effective communication, public test-ride events, and transparency in safety measures as pivotal factors in shaping CAV social acceptability and strengthening cybersecurity, as evidenced by CAV users' and OEMs' education scenarios. The CAV Penetration with Public Behaviour Analysis scenario further emphasises the value of understanding public perceptions and behaviours to improve CAV cybersecurity resilience. Importantly, the findings identify humans as a critical component within the security framework of CAVs, often referred to as the "weakest link," underscoring the urgent need for enhanced public awareness and education. While much of the existing literature focus remains on technical measures and the categorisation of security actors, this study advocates for a significant shift toward recognising and emphasising the role of the general public as an integral part of the CAV cybersecurity ecosystem, where human involvement plays a crucial role in mitigating risks and enhancing the security of autonomous vehicles. A new policy horizon lies in equipping individuals with the awareness, knowledge, and practical skills to mitigate security risks associated with CAVs. Public education campaigns, user-friendly guidelines, and targeted training initiatives can empower individuals to adopt safer digital practices, recognise potential threats, and contribute to a secure CAV environment. By integrating the general public into the broader cybersecurity strategy, policymakers can create secure CAV systems that leverage the collective vigilance of informed users. Similarly, based on these scenario findings, the study advocates for the implementation of strategic measures by vendors and regulators to strengthen cybersecurity and improve the resilience of the CAV ecosystem.

CRedit authorship contribution statement

Shah Khalid Khan: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nirajan Shiwakoti:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Peter Stasinopoulos:** Writing – review & editing, Supervision, Methodology, Investigation, Formal

analysis, Conceptualization. **Yilun Chen:** Formal analysis, Conceptualization. **Matthew Warren:** Supervision.

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Appendix A. The model parameters and expressions

Name	Expression	Unit
Adoption from Imitation	$Imitation * CAVs\ Adopters * CV\ Adopters / Total\ Car\ Adaptors$	Vehicle/Year
Adaption from Innovation	$CV\ Adopters * Innovation$	Vehicle/Year
Awareness Enhancement	$(CAVs\ User\ and\ OEMs\ Education\ Effectiveness * Criminology\ Theory\ Maturity)$	Dmnl/Year
Capability Depreciation	$Hacker's\ Capability / Capability\ Lifetime$	Dmnl/Year
Probability of Hacks Defended	$0.6 * CAVs\ Communication\ Cyber\ Safety\ Stack + 0.15 * Human\ CAV\ Cyber\ Safety\ Awareness + 0.25 * CAVs\ Communication\ Cyber\ Safety\ Stack * Human\ CAV\ Cyber\ Safety\ Awareness$	Dmnl
Adoption	$(Adaption\ from\ Innovation + Adoption\ from\ Imitation) * (1 - Probability\ of\ Successful\ Cyber\ attacks)$	Vehicle/Year
Awareness Lifetime	50	Year
Awareness Depreciation	$Human\ CAV\ Cyber\ Safety\ Awareness / Awareness\ Lifetime$	Dmnl/Year
Capability Lifetime	50	Year
Capability per Log File	$Capability\ per\ Leaked\ Log\ File * Leaked\ Log\ Files$	Dmnl
Capability per Vehicle	0.5 on a {0–1} scale	Dmnl/Vehicle
CAVs Adopters	$CAVs\ adopters\ Initial + \sum_0^t (-Adoption + Induction) \Delta t$ CAVs adopters Initial = 0.001	Vehicle
CAVs Communication Cyber Safety Stack	$CAVs\ Communication\ Cyber\ Safety\ Stack\ Initial + \sum_0^t (Robustness\ Enhancement - Knowledge\ \&\ Technology\ Depreciation) \Delta t$ CAVs Communication Cyber Safety Stack Initial = 0	Dmnl
CAVs Infiltration Caveats	$Capability\ per\ Vehicle * CAVs\ Adopters$	Dmnl
CAVs Technological Maturation	$Information * Processes * Resources * Research\ and\ Development * CAVs\ Technology\ Readiness\ and\ Maturity * Initial\ Maturation\ Availability$	Dmnl/Year
CAVs Technology Readiness and Maturity	$CAVs\ Technology\ Readiness\ and\ Maturity\ Initial + \sum_0^t (CAVs\ Technological\ Maturation) \Delta t$ CAVs Technology Readiness and Maturity Initial = 0.1 Non-zero initial value assumption	Dmnl
CAVs User and OEMs Education Effectiveness	0.5 on a {0–1} scale	Dmnl/Year
Criminology Theory Maturation	$eSafety\ Traffic\ Unit * (Human\ (User)\ Behaviour\ Analysis + Knowledge\ of\ CAV\ Attackers' \ Motivations\ and\ Characteristics)$	Dmnl/Year
Criminology Theory Maturity	$Criminology\ Theory\ Maturity\ Initial + \sum_0^t (Criminology\ Theory\ Maturation) \Delta t$ Criminology Theory Maturity Initial = 0	Dmnl
CV Adopters	$CV\ adopters\ Initial + \sum_0^t (-Adoption) \Delta t$	Vehicle
CV adopters Initial	Population * Ownership initial - CV always - AV adopters initial (Population = 1, Ownership initial 0.57, CV always = 0.1, AV adopters initial = 0.0001)	Vehicle
Deployment Rate	0.5	1/Year
eSafety Traffic Unit	0.5 on a {0–1} scale	Dmnl
Hacker's Motivation	0.5 on a {0–1} scale	Dmnl/Year
Hacker's Technical advancement	$CAVs\ Communication\ Cyber\ Safety\ Stack * Hackers\ Accessibility$	Dmnl
Hacker's Accessibility	0.5 on a {0–1} scale	Dmnl
Hacker's Capability	$Hacker's\ Capability\ Initial + \sum_0^t (Hackers\ Capability\ Enhancement - Capability\ Depreciation) \Delta t$ Hacker's Capability Initial = 0	Dmnl
Hackers Capability Enhancement	$(CAVs\ Infiltration\ Caveats + Capability\ per\ Log\ File + Hacker's\ Technical\ advancement) * Hacker's\ Motivation$	Dmnl/Year
Human (User) Behaviour Analysis	$CAVs\ Adopters * Knowledge\ per\ User$	Dmnl/Year
Human CAV Cyber Safety Awareness	$Human\ CAV\ Cyber\ Safety\ Awareness\ Initial + \sum_0^t (CAVs\ Technological\ Maturation) \Delta t$ Human CAV Cyber Safety Awareness Initial = 0.1	Dmnl
Imitation	0.1	1/Year
Induction from Imitation	$Imitation * CAVs\ Adopters * Non-Car\ Adopters / Total\ Non-car\ Adopters$	Vehicle/Year
Induction	$(Induction\ from\ Imitation + Induction\ from\ Innovation) * (1 - Probability\ of\ Successful\ Cyber\ attacks)$	Vehicle/Year
Induction from Innovation	$Innovation * Non-Car\ Adopters$	Vehicle/Year
Information	$Log\ Files\ Preservation * Information\ per\ Log\ File$	Dmnl/Year
Information per Log File	0.4 on a {0–1} scale	Dmnl/File
Innovation	$CAVs\ Technology\ Readiness\ and\ Maturity * Innovation\ per\ Readiness$	1/Year
Initial Maturation Availability	$Initial\ Maturation\ Availability\ Initial - \sum_0^t (CAVs\ Technological\ Maturation) \Delta t$ Initial Maturation Availability Initial = 0.9	Dmnl
Input Log Files	1	File/Year
Knowledge & Technology Depreciation	$CAVs\ Communication\ Cyber\ Safety\ Stack / Knowledge\ \&\ Technology\ Lifetime$	Dmnl/Year
Knowledge & Technology Lifetime	20	Year
Knowledge of CAV Attackers' Motivations and Characteristics	$Log\ Files\ Preservation * Knowledge\ per\ Log\ File$	Dmnl/Year
Knowledge per Log File	0.001	Dmnl/File
Knowledge per User	0.5 on a {0–1} scale	Dmnl/Vehicle/Year
Leakage rate	0.0025	1/Year
Leaked Log Files	$0 + \int (Leaking) \Delta t$	File
Leaking	$Log\ Files * Leakage\ rate$	File/Year
Log Files	$Log\ Files\ Initial + \sum_0^t (Log\ Files\ Preservation - Leaking) \Delta t$ Log Files Initial = 0.2	File

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Name	Expression	Unit
Learning Experiences	0.3 * Probability of Successful Cyber-attacks	Dmnl
Log Files Preservation	Input Log Files *Learning Experiences	File/Year
Non-Car Adopters	CAVs adopters Initial + $\sum_0^t (- Adoption + Induction) \Delta t$	Vehicle
Non-car adopters Initial	Non-car adopters initial * Induction Scenario Induction Scenario = 1 (yes) Induction Scenario = 0 (No)	Vehicle
Non-car adopters initial	Population * (1 - Ownership initial)	Vehicle
Probability of Hacking attempts	Hacker's Capability * (1 - Criminology-Theory Maturity)	Dmnl
Probability of Successful Cyber-attacks	Probability of Hacking attempts * (1- Probability of Hacks defended)	Dmnl
Processes	0.5 on a {0–1} scale	Dmnl
Research and Development	0.5 on a {0–1} scale	Dmnl
Resources	0.5 on a {0–1} scale	Dmnl
Robustness Enhancement	V2X Communication Security * Deployment Rate	Dmnl/Year
Total Car Adaptors	CV Adaptors + CAVs Adaptors	Vehicle
Total Non-car Adaptors	CAVs Adaptors + Non-Car Adaptors	Vehicle

Data availability

Data will be made available on request.

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